

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 개념_*.pdf(슬라이드/치트시트)로 핵심 공식을 먼저 최종 점검한 뒤 풀이 시작

문항 목차 (총 12문항)

- [1] L05 · IPSRP 4.4.15 — 연속RV1
- [2] L05 · PSP 4.5.16 — 연속RV1
- [3] L06 · IPSRP 5.4.22 — 연속RV2
- [4] L06 · PSP 5.6.6 — 연속RV2
- [5] L07 · IPSRP 5.4.34 — Further1 변환·공분산
- [6] L07 · PSP 6.4.1 — Further1 변환·공분산
- [7] L07 · PSP 6.5.1 — Further1 변환·공분산
- [8] L09 · PSP 10.1.2 — 점근 CLT
- [9] 복습 L02 · PSP 1.4.5 — 조건부·Bayes
- [10] 복습 L03 · PSP 3.9.1 — 이산RV1
- [11] 복습 L04 · PSP 5.2.2 — 이산RV2
- [12] 복습 L04 · PSP 5.3.2 — 이산RV2

Problem 15

Let X be a mixed random variable with the following generalized PDF

$$f_X(x) = \frac{1}{3}\delta(x+2) + \frac{1}{6}\delta(x-1) + \frac{1}{2}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}.$$

- a. Find $P(X = 1)$ and $P(X = -2)$.
- b. Find $P(X \geq 1)$.
- c. Find $P(X = 1|X \geq 1)$.
- d. Find EX and $Var(X)$.

4.5.16■ In this problem we verify that an Erlang (n, λ) PDF integrates to 1. Let the integral of the n th order Erlang PDF be denoted by

$$I_n = \int_0^\infty \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!} dx.$$

First, show directly that the Erlang PDF with $n = 1$ integrates to 1 by verifying that $I_1 = 1$. Second, use integration by parts (Appendix B, Math Fact B.10) to show that $I_n = I_{n-1}$.

Problem 22

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} \frac{1}{2}x^2 + \frac{2}{3}y & -1 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[Y|X = 0]$ and $\text{Var}(Y|X = 0)$.

5.6.6 ● X_1 and X_2 are independent random variables such that X_i has PDF

$$f_{X_i}(x) = \begin{cases} \lambda_i e^{-\lambda_i x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

What is $P[X_2 < X_1]$?

Problem 34

Let $X \sim \text{Uniform}(1, 3)$ and $Y|X \sim \text{Exponential}(X)$. Find $\text{Cov}(X, Y)$.

— 풀이 —

6.4.1● Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6xy^2 & 0 \leq x, y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $V = \max(X, Y)$. Find the CDF and PDF of V .

6.5.1 ● Let X and Y be independent discrete random variables such that $P_X(k) = P_Y(k) = 0$ for all non-integer k . Show that the PMF of $W = X + Y$ satisfies

$$P_W(w) = \sum_{k=-\infty}^{\infty} P_X(k) P_Y(w - k).$$

10.1.2 ● $X_1, \dots, X_n$ are independent uniform random variables with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$.

- (a) What is the PDF of X_1 ?
- (b) What is $\text{Var}[M_{16}(X)]$, the variance of the sample mean based on 16 trials?
- (c) What is $P[X_1 > 9]$, the probability that one outcome exceeds 9?
- (d) Would you expect $P[M_{16}(X) > 9]$ to be bigger or smaller than $P[X_1 > 9]$?

To check your intuition, use the central limit theorem to estimate $P[M_{16}(X) > 9]$.

1.4.5 ■ The basic rules of genetics were discovered in mid-1800s by Mendel, who found that each characteristic of a pea plant, such as whether the seeds were green or yellow, is determined by two genes, one from each parent. In his pea plants, Mendel found that yellow seeds were a dominant trait over green seeds. A yy pea with two yellow genes has yellow seeds; a gg pea with two recessive genes has green seeds; a hybrid gy or yg

pea has yellow seeds. In one of Mendel's experiments, he started with a parental generation in which half the pea plants were yy and half the plants were gg . The two groups were crossbred so that each pea plant in the first generation was gy . In the second generation, each pea plant was equally likely to inherit a y or a g gene from each first-generation parent. What is the probability $P[Y]$ that a randomly chosen pea plant in the second generation has yellow seeds?

— 풀이 —

3.9.1 ● Let X be the binomial $(100, 1/2)$ random variable. Let E_2 denote the event that X is a perfect square. Calculate $P[E_2]$.

5.2.2 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} c|x+y| & x = -2, 0, 2; \\ & y = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c ?

(b) What is $P[Y < X]$?

(c) What is $P[Y > X]$?

(d) What is $P[Y = X]$?

(e) What is $P[X < 1]$?

5.3.2● Given the random variables X and Y in Problem 5.2.2, find

- (a) The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- (b) The expected values $E[X]$ and $E[Y]$,
- (c) The standard deviations σ_X and σ_Y .